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Beyond Direct Economic Losses: An Integrated Approach to Seismic Retrofit Considering Sustainability and Indirect Losses

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ABSTRACT

Seismic retrofit stands as a cost-effective method to strengthen vulnerable buildings and enhance the resilience of earthquake-prone communities. The selection of suitable retrofit strategies is often facilitated by decision-support schemes, which enable the comparison of various options against user-defined criteria (weighted according to stakeholder preferences). Traditional approaches, however, have predominantly emphasised monetary metrics like the costs of retrofit intervention and seismic damage repair, while overlooking other pivotal factors for more informed decision making, including environmental sustainability and indirect economic losses due to downtime. Furthermore, retrofit strategies are rarely designed to achieve the same performance objectives or risk targets before being evaluated via decision-support schemes, possibly leading to misguided comparisons and non-optimal choices. This study addresses these gaps by examining the role of indirect losses on retrofit feasibility and design by incorporating their risk-reduction benefit into conventional life-cycle cost analysis under varying seismic hazard levels. The study also explores prospective pathways for integrating sustainability metrics into decision-making by assessing the carbon footprints of buildings with a life-cycle lens, in addition to the application of carbon taxes. A case study is presented involving an older reinforced concrete (RC) frame, retrofitted using three common strategies: fibre-reinforced polymer wrapping of columns/joints, RC jacketing, and steel jacketing. Finally, the study proposes a practitioner-friendly approach for retrofit design optimisation with minimal iterations. It allows users to specify their desired seismic risk level (in terms of losses) and returns the corresponding nominal performance that must be targeted during the design process.

1 | Introduction

Recent devastating earthquakes [1, 2] have underlined the urgent need for seismic risk mitigation measures to enhance community resilience. This is particularly crucial in regions with numerous non-code-compliant buildings that are prone to extensive damage, or even collapse, under moderate-to-high ground shaking. While demolishing these buildings to replace them with code-compliant ones would effectively reduce seismic vulnerability, such an approach is often financially unfeasible and environmentally detrimental. Instead, structural retrofitting of existing buildings to improve their seismic performance emerges as a practical and more eco-friendly solution. Retrofit strategies can involve global interventions like adding shear walls/bracing [3, 4] or local enhancement of weak structural members, such as encasing columns with reinforced concrete (RC) or steel

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jackets, or wrapping them with fibre-reinforced polymers (FRP) to increase their strength and ductility.

A critical aspect of structural retrofit is selecting the most suitable strategy that upgrades a building's seismic response while remaining economically feasible. This task becomes more intricate when applied to building portfolios, where budget constraints may substantially hinder large-scale retrofit initiatives. Several approaches have been proposed in the literature to tackle this issue. For instance, Liel and Deierlein [5] conducted a benefit-cost analysis (BCA) to assess the retrofit effectiveness for older RC frames, weighing the reduction in direct losses (i.e., repair costs) and casualties against the cost of retrofit intervention. Other forms of BCA, like breakeven times, have also been adopted (e.g., [6]). Although BCA is a valuable instrument for assessing retrofit feasibility, it falls short in optimising/comparing different alternatives. A retrofit strategy with a larger economic benefit-cost ratio, for example, is not necessarily the optimal solution minimising a building's life-cycle cost.

To overcome the above limitations, some studies offered algorithms that optimise retrofit layouts/designs based on specific performance or risk targets [7, 8]. Others advocated for multi-criteria decision-making (MCDM) methods, which identify and rank retrofit alternatives against user-defined criteria, considering their relative importance (weights). MCDM can also accommodate qualitative criteria (e.g., architectural compatibility, disruption). One early MCDM application by Caterino et al. [9] compared common retrofit strategies for a case-study building. However, their retrofit design process did not ensure similar seismic performance across retrofit strategies, leading the structure to experience varying damage levels, even under the same ground-shaking intensity. This issue was addressed by Muntasir Billah and Shahria Alam [10] and Gentile and Galasso [11], who designed their retrofits to achieve comparable performance at the target seismic hazard prior to applying MCDM. A similar approach was later carried out by Gallo et al. [12].

While selecting the best retrofit strategy among multiple alternatives has been extensively examined in the literature, the economic consequences of earthquakes in decision-making are typically limited to direct losses associated with repairing physical damage. Conversely, indirect losses incurred due to downtime and business interruption, which can exceed direct losses, have received limited attention [13, 14]. Among the few studies addressing this issue is Caruso et al. [15], but they focused solely on a single retrofit strategy. This highlights the need to further explore the influence of indirect losses and their risk-reduction benefit when comparing different retrofit strategies.

An additional challenge arises from the growing global recognition of building sustainability in seismic zones and the environmental impacts induced by earthquakes, such as greenhouse gases that contribute to climate change. These impacts stem from the processes needed to repair damaged building components, including material production, transportation, restoration, and demolition/disposal [16, 17]. Clemett et al. [18] was one of the few studies to integrate repair costs and environmental impacts of seismic damage via MCDM to evaluate retrofit options, suggesting that those with the least repair costs have also the lowest environmental impacts, given the strong correction between the two metrics. This led Caruso et al. [19, 20] to propose combining structural retrofits with energy upgrades (e.g., thermal insulation) to further enhance life-cycle sustainability. While this approach made environmental impacts more influential in MCDM, energy upgrades can significantly increase intervention costs [21], potentially limiting their feasibility in large-scale retrofit plans. This leaves a series of critical questions: *should earthquake-induced environmental impacts serve as a criterion to compare retrofit strategies? Is it reasonable to assume that the retrofit strategy offering the greatest reduction in repair costs is, in parallel, the most sustainable?*

Lastly, most past studies compared retrofit strategies that were designed based on engineering judgment rather than pre-defined performance objectives or risk targets. This implies that any alteration in the layout/geometric details of one strategy can entirely change decision-making outcomes. Ideally, each retrofit strategy must be individually designed and optimised before being compared with other options to ensure informed choices. Such optimisation requires an iterative process that inspects a wide range of retrofit layouts/geometries, ultimately selecting the one meeting a targeted risk level or offering an optimal balance between investment and risk reduction (e.g., minimal life-cycle costs). This process is often avoided due to its computational expense, especially at a building-portfolio level. Thus, a straightforward, practitioner-friendly approach is necessary to guide and streamline the design optimisation with fewer iterations.

This study addresses the identified research gaps via three pillars: (1) explore pathways to integrate life-cycle sustainability into decision-making of seismic retrofit, considering the environmental impacts of both the retrofit intervention and seismic damage. This enables understanding their influence on decision outcomes, compared to conventional, highly correlated metrics like repair and life-cycle costs; (2) assess the effect of indirect losses on decision-making by estimating their risk-reduction benefit across multiple hazard levels; and (3) propose a practice-oriented approach that guides retrofit design to achieve a targeted risk-reduction level or minimal life-cycle costs with reduced effort. This ensures that each retrofit strategy is individually optimised before comparison via decision-making schemes, facilitating more informed decision outcomes. These pillars are demonstrated via an archetype older RC frame retrofitted using three widespread strategies: FRP wrapping of columns/joints, steel jacketing, and RC jacketing. Multiple retrofit layouts/configurations are adopted for each strategy to capture variations in design choices and their impact on economic and sustainability metrics. Simple models (i.e., equations) are also derived from empirical data and simulation to compute both the environmental impact and cost of a retrofit intervention based on its geometry. This allows one to compare retrofit strategies based on their life cycle cost and sustainability, instead of relying solely on seismic-damage consequences.

2 | Methodology

The workflow adopted in this study, depicted in Figure 1, encompasses the following steps: (1) identify a case-study structure; (2–3) design retrofit strategies with various layouts and geometries, and simultaneously develop numerical models to simulate

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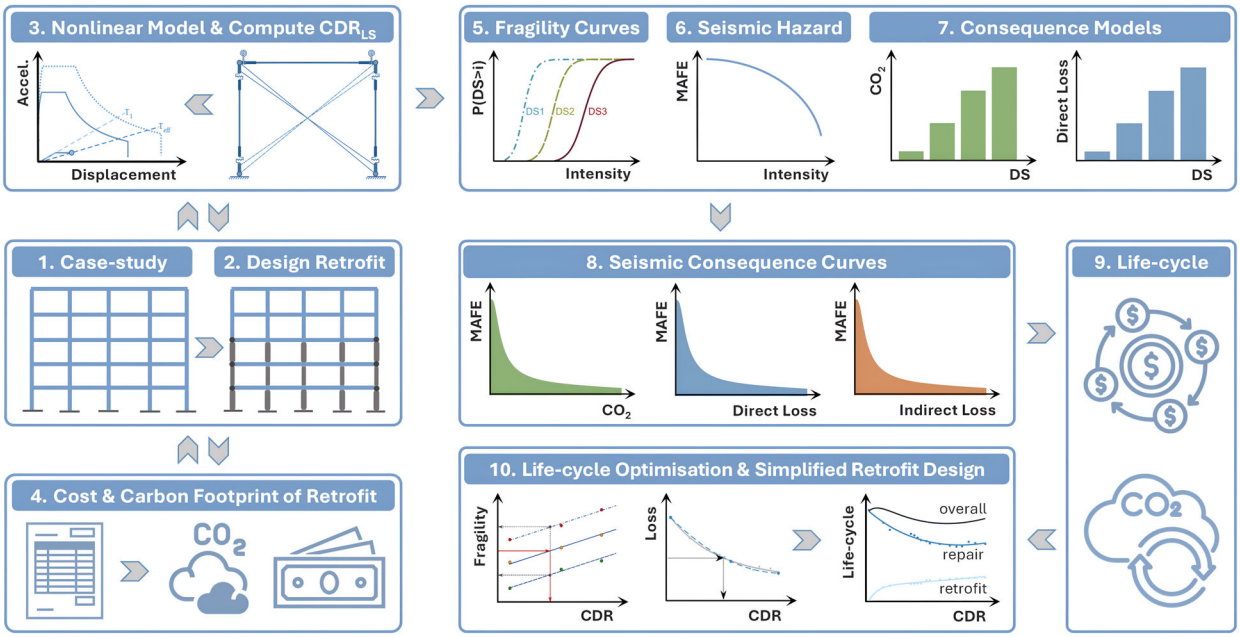


FIGURE 1 | A schematic representation of the workflow adopted in this study.

the non-linear behaviour; (4) evaluate the (initial) costs and environmental impacts of each retrofit configuration; (5) derive fragility relationships defining the exceedance probabilities of different damage levels (or states) for various ground-motion intensity levels; (6) characterise site-specific seismic hazard; (7) select consequence models linking the considered damage states to a consequence metric of interest; (8) generate seismic consequence curves; (9) obtain the life-cycle consequences of seismic damage, including repair costs, environmental impacts, and indirect losses, to explore their effect on optimal retrofit decision-making; and finally (10) use the results to discuss a practice-oriented approach for optimising the design of any retrofit strategy with minimal iterations.

2.1 | Case-Study Building, Retrofit Strategies, and Non-Linear Modelling

This study considers an older five-storey RC frame, 15 m in height, with four 4.5 m bays in both horizontal directions (Figure 2a), which was designed by Aljawhari et al. [22] for gravity loads only to reflect buildings constructed in Italy before the 1970s. The case-study frame exhibits poor performance under earthquake-induced ground shaking due to the absence of seismic provisions such as confinement and capacity design. To address these issues, three retrofit strategies are selected (Figure 2b): (1) FRP wrapping of joints and columns using pre-cured sheets of carbon fibres to prevent shear failure and improve ductility via lateral confinement [23]; (2) steel jacketing of columns via full-height elliptical jackets that provide similar benefits to FRP; (3) RC jacketing of columns using cast-in-situ RC encasements, which improve confinement, ductility and shear and flexural strengths. The jacket's thickness is governed by the rebar's size and required minimum cover. Material and geometric details for the three strategies are available in Aljawhari et al. [22].

The non-linear response of the as-built and retrofitted frames is simulated using 2D numerical models developed in OpenSees [24] in an earlier work by the authors [22]. Structural components were modelled as beam-column elements with finite-length plastic hinges to capture flexural responses, with shear springs added in series to tackle shear failure. Joint failure was explicitly modelled. The effects of FRP wrapping and steel jacketing were incorporated by modifying column curvature ductility and adding their contribution to shear strength. RC-jacketed columns, on the other hand, were treated as equivalent monolithic members. Infill walls were not modelled to mimic prevailing design practices for both retrofitted buildings and new constructions. Further details on the modelling approach can be found in Aljawhari et al. [22].

2.2 | Design of Retrofit Strategies

Aljawhari et al. [22] developed numerous configurations for each retrofit strategy via an incremental design procedure. This process systematically varied detailing, geometric features, and the number of retrofitted structural elements to achieve a gradual improvement in seismic performance. The improvement was, in turn, captured by estimating the ratio between the building's global displacement capacity and life-safety seismic demand (CDR_{LS}), or simply the capacity-demand ratio. This metric was introduced under the name 'percentage of new building standard (%NBS)' in the guidelines for seismic assessment of existing buildings by the New Zealand Society for Earthquake Engineering [25] and further used in several studies (e.g., [22, 26]). A similar metric exists in the Italian Guidelines for Seismic Risk Classification [27]. The life-safety seismic demand was defined as a Type-1 response spectrum from Eurocode 8 [28], with a peak ground acceleration of 0.30 g (475-year return period) and ground type C. The CDR_{LS} value for each retrofitted configuration was computed through non-linear static (pushover) analysis in conjunction with

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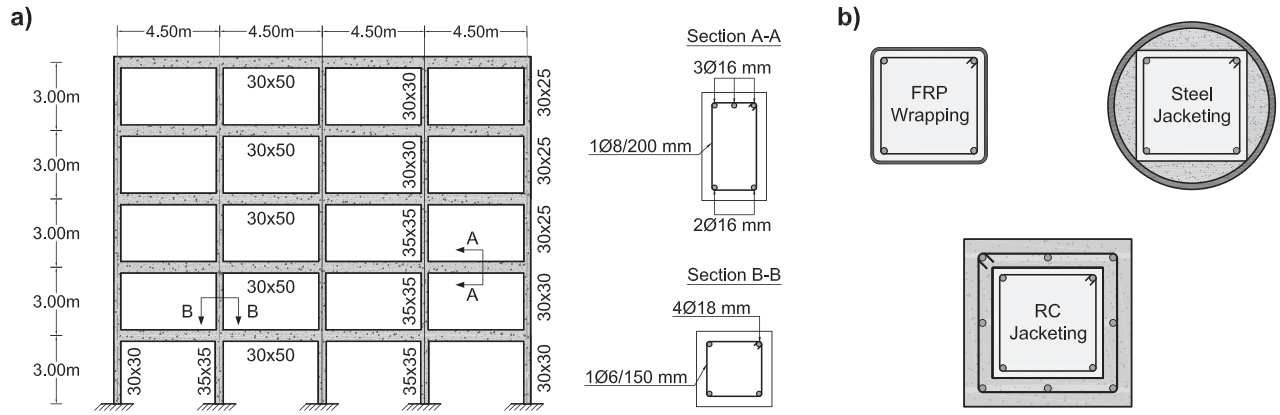


FIGURE 2 | As-built case-study frame: (a) geometry/reinforcement; and (b) adopted retrofit strategies.

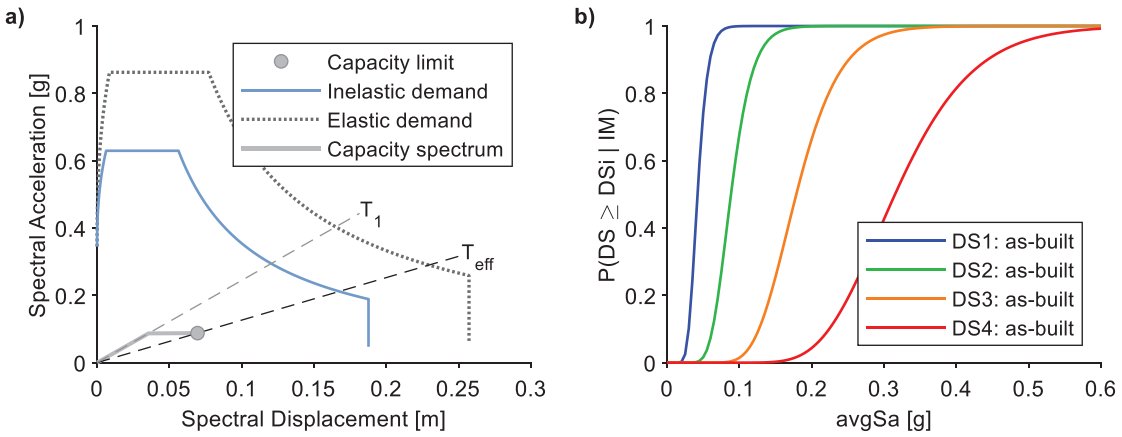


FIGURE 3 | As-built case-study frame: (a) evaluation of CDR_{LS} using the capacity spectrum method; and (b) fragility relationships.

the capacity spectrum method [29]. Figure 3a shows how the demand spectrum significantly exceeds the capacity spectrum, yielding a CDR_{LS} of just 42%.

The incremental design process yielded 13 retrofitted configurations using FRP wrapping, 15 for steel jacketing, and 17 for RC jacketing. Table 1 reports the layout and geometric details of select configurations, with the remaining ones described in Aljawhari et al. [22]. The largest CDR_{LS} value achieved via FRP wrapping and steel jacketing is 110% and 147%, respectively. RC jacketing reached a CDR_{LS} of 181% as is the only strategy that enhances both ductility and lateral strength, shifting the plastic behaviour to a global beam-sway mechanism.

2.3 | Derivation of Fragility Relationships

Computing seismic consequences for both the as-built and retrofitted structures begins with characterising earthquake damage levels, or damage states (DSs), through fragility relationships, which define the probability of exceeding a specific DS at different levels of a ground-motion intensity measure (IM), that is, $P(DS \geq DSi|IM)$. These relationships are modelled as cumulative lognormal distributions with a median, μ_{DSi}, and a dispersion, β, for the 'i-th' DS (DSi). The adopted IM is the geometric mean of the 5%-damped spectral acceleration values over a defined range of vibration periods (avgSa), which accounts for higher-mode effects and period elongation due to stiffness degradation [30]. Four DSs are considered: DS1 represents minor non-structural cracks; DS2 involves moderate structural and non-structural damage without significant yielding; DS3 corresponds to extensive damage; and DS4 reflects near-collapse conditions with fully exploited ductility and deficient residual strength. The occurrence of each DS is inferred numerically through structural analysis by estimating the maximum interstorey drift ratio (MIDR) due to seismic action and then comparing it with the corresponding MIDR thresholds marking the onset of each DS. These thresholds are calibrated by assessing the multi-level damage criteria defined in Aljawhari et al. [31] using pushover analysis.

Fragility relationships for DS2 through DS4 were derived by Aljawhari et al. [22] for the as-built and retrofitted configurations via the cloud-analysis-based procedure of Jalayer et al. [32]. The same procedure is used here to derive DS1 fragility relationships (see an example in Figure 3b). Figure 4 illustrates the variation of CDR_{LS} and the corresponding μ_{DSi} across all retrofitted configurations, highlighting the incremental improvement in seismic performance. Gradual increases in μ_{DS3} and μ_{DS4} are observed for all retrofit strategies. However, μ_{DS1} and μ_{DS2} increases are only pronounced for RC jacketing as it significantly enhanced the lateral strength and stiffness. The β values for all fragility sets are consistent, ranging from 0.24 to 0.29. A more in-depth discussion is available in Aljawhari et al. [22].

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TABLE 1 | Description of some retrofitting configurations using the three strategies (grey lines: retrofitted columns, black circles: retrofitted joints).

Retrofit	CDR_{LS}	Description of the retrofit design/layout/geometry	Illustration
FRP wrapping	65%	1 layer for exterior columns and joints in the 1st to 3rd floor and 1 layer for interior columns in the 1st and 3rd floors	
	105%	Same joint retrofit as above. 1 layer for the 2nd-floor columns. 4 layers for the 1st-floor and interior columns on the 3rd floor and 5 layers for the 3rd-floor exterior columns	
Steel jacketing	61%	1.5 mm thick jacket for the exterior columns on the 1st to 3rd floor and interior ones on the 3rd floor. Retrofit exterior joints in the 1st to 3rd floors using FRP	
	106%	Same joint retrofit as above. 1.5 mm thick jacket for the columns of the 2nd floor. 2.5 mm thick jackets for the 3rd floor and exterior columns of the 1st floor. 3 mm jacket for the interior columns on the 1st floor	
RC jacketing	65%	Retrofit exterior columns on the 1st to 3rd floor with 50 mm jacket with 4Φ14 mm bars and 150 mm-spaced hoops	
	105%	50 mm jackets for the exterior columns on the 1st to 4th floor with 4Φ14 mm bars and 150 mm-spaced hoops. 50 mm jackets for the interior columns in 1st, 3rd, 4th floors with 4Φ16 mm bars and 150 mm-spaced hoops	

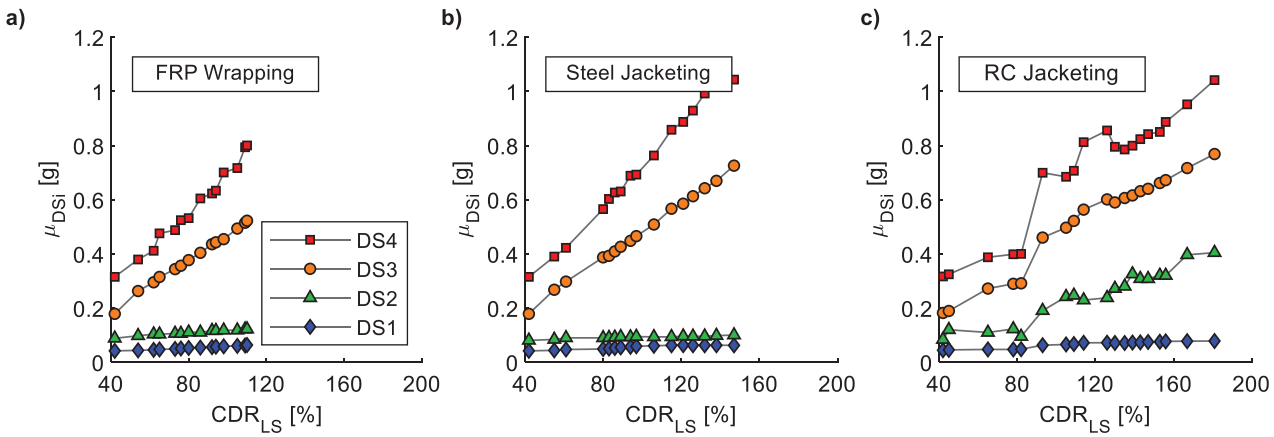


FIGURE 4 | Variation of CDR_{LS} versus μ_{DSi} for (a) FRP wrapping; (b) steel jacketing; and (c) RC jacketing.

2.4 | Site-Specific Seismic Hazard

To address earthquake consequences under high seismicity, it is assumed that the case-study frame is located in L'Aquila, Italy (42.350 N°, 13.400 E°) and that the soil shear-wave velocity is 270 m/s. Hazard analysis is then performed via the OpenQuake engine [33], adopting the 2013 Euro-Mediterranean Seismic Hazard Model [34] and the assumptions of Aljawhari et al. [31]. Another Italian city is considered to account for low-seismicity conditions (Florence, 43.813 N°, 11.204 E°). Figure 5 presents the

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TABLE 2 | Adopted DLRs for different DSs and design-code levels.

Structure/DS	DS1 DLR1 [%]	DS2 DLR2 [%]	DS3 DLR3 [%]	DS4 DLR4 [%]
5-storey pre-code	10.9	32.2	62.2	84.6
5-storey low-code	11.3	35.5	63.8	87.2
5-storey high-code	12.9	36.5	68.4	88.7

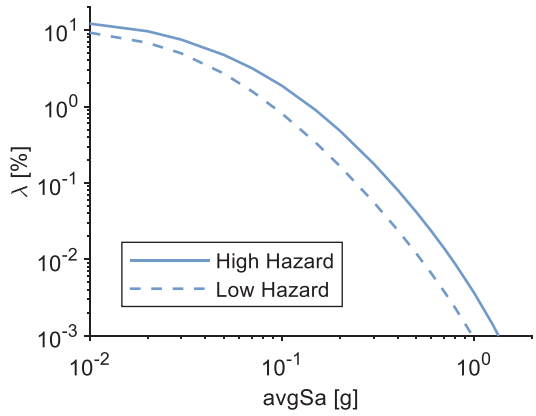


FIGURE 5 | Hazard curves for the high (L'Aquila) and low (Florence) seismicity conditions.

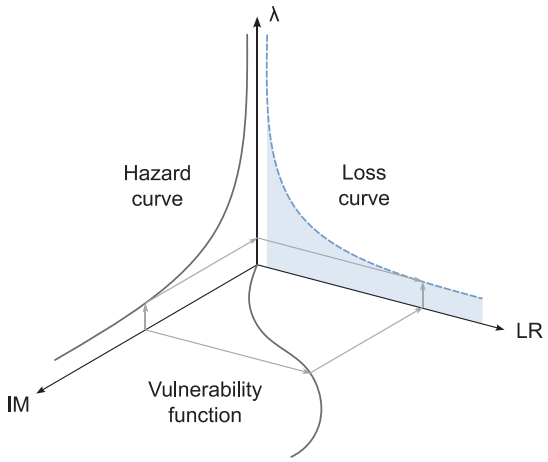


FIGURE 6 | Deriving the loss curve by combining the vulnerability function with the hazard curve.

hazard curves characterising the range of IM (i.e., avgSa) levels of interest and their mean annual exceedance frequency (λ).

2.5 | Earthquake-Induced Repair Costs

The expected repair costs (direct losses) of seismic damage at different IM levels can be quantified through vulnerability functions, derived as per Equation (1) by combining the DS probabilities from fragility relationships with suitable damage-to-loss ratios (DLRs). DLRs represent the repair costs incurred when a building attains specific global DSs normalised by its replacement cost (*ReC*). Accordingly, vulnerability functions express the loss as a ratio (*LR*) of the repair costs to *ReC*; *n* refers to the number of DSs. The DLRs adopted here were numerically developed by Aljawhari et al. [31] for archetype Italian RC frames via a simulation-based procedure. Table 2 reports the mean values of these DLRs for the four DSs.

$$LR\,(IM) = \sum_{i=1}^{n} P\,(DS = DSi|IM)\,DLRi \tag{1}$$

Vulnerability functions are subsequently combined with hazard data to generate consequence (i.e. loss) curves, which characterise the λ values for a range of *LRs* (see Figure 6). Integrating these loss curves as per Equation (2) yields the expected annual loss (EAL), a crucial seismic risk metric that reflects the average annual repair costs a building is likely to incur due to earthquake damage, considering the entire spectrum of possible ground-shaking scenarios. This metric is essential for examining the life-cycle financial implications of seismic risk, as explained later.

For practical application, the EAL in Equation (2) is computed as a ratio relative to the building's *ReC*.

$$EAL = \int_{0}^{\infty} LR\,(\lambda)\,d\lambda \tag{2}$$

2.6 | Earthquake-Induced Environmental Impacts

The environmental impacts resulting from the repair activities of earthquake damage, such as material production, construction, and demolition/disposal activities, are quantified in terms of embodied carbon, a widely-recognised metric for assessing global warming potential, and its contribution to climate change [35]. It calculates the total greenhouse gas emissions generated by the previous processes in equivalent kilograms (kgs) of carbon dioxide (CO_2e). For simplicity, the term 'embodied carbon' is referred to as just 'carbon' from this point onwards.

A practical method for estimating the carbon emissions of repairing a building component involves identifying the required type and quantity of construction materials, depending on the damage level. These quantities are multiplied by the so-called carbon factors characterising the emissions of processes like material production, transportation, or disposal per material unit quantity. Carbon factors are typically derived from empirical or manufacturer-provided environmental data, and can be accessed through many databases [36, 37]. This method should be applied to all damaged building components to obtain the total carbon emissions.

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TABLE 3 | Adopted DCRs for different DSs and design-code levels.

Structure/DS	DS1 DCR1 [%]	DS2 DCR2 [%]	DS3 DCR3 [%]	DS4 DCR4 [%]
5-storey pre-code	8.3	32.5	60.2	80.2
5-storey low-code	8.8	41.1	69.2	87.1
5-storey high-code	11.9	43.1	73.5	90.2

Alternatively, carbon emissions of seismic-damage repair can be estimated at the building's level, which requires damage-to-carbon ratios (DCRs). DCRs translate the global DSs experienced by a building to the corresponding carbon emissions, normalised by its replacement carbon ($ReCO_2$). Vulnerability functions are then developed to determine the carbon ratio (CR) at different IM levels by combining the DS probabilities with suitable DCRs, as per Equation (3). The mean DCR values, outlined in Table 3, are adopted from the authors' previous work on deriving numerical carbon consequence models utilising a simulation-based procedure and empirical environmental-impact data [38]. Once the CR functions are derived, they are combined with seismic hazard to acquire environmental consequence curves defining λ for any CR value (see Figure 6). These curves can be integrated via Equation (4) to compute the expected annual carbon (EAC), which captures the average yearly carbon emissions of repairing seismic damage (normalised by $ReCO_2$).

$$CR(IM) = \sum_{i=1}^{n} P(DS = DSi|IM)\, DCRi \tag{3}$$

$$EAC = \int_0^{\infty} CO_2R(\lambda)\, d\lambda \tag{4}$$

2.7 | Earthquake-Induced Indirect Losses

Indirect loss arises from business interruptions due to building downtime, and consists of two key components: (1) a 'rational' component, representing the time required to accomplish repair works (i.e., repair time) and (2) an 'irrational' component covering delays before the initiation of repairs, like those due to bidding, design, and financing [39]. Various approaches exist for computing downtime. For instance, Mitrani-Reiser [40] estimated it based on buildings' collapse potential, while Molina Hutt et al. [41] developed a framework that incorporates the recovery state of the building, considering mobilisation delays, repair time and utility disruption. However, their framework deployed factors tailored to the US context and may not accurately reflect the situation in Italy. Moreover, they used component-based seismic loss assessments, whereas this study evaluates loss at the building level, which is more suitable for building portfolios.

In residential buildings, indirect losses primarily stem from the costs of relocating occupants to temporary housing during repairs. To evaluate these losses, the current study adopts the simplified procedure of Calvi et al. [42], which multiplies the daily cost of relocation per person (C_{rel}) by the number of occupants (N_p) and the total downtime. The N_p is estimated assuming that each person needs a space of 25 m^2 [42], knowing that the minimum space per person is 14 m^2 as per the July 5th, 1975, decree by the Italian Ministry of Health [43]. It is assumed here that relocation is necessary solely when the building suffers or exceeds extensive damage ($\geq$ DS3).

Equation (5) outlines the derivation for the vulnerability functions in terms of the mean indirect loss ratio (LR_{ind}), given the ground-shaking IM. DTRi is the damage-to-repair time ratio expressing the time needed to repair the 'i-th' DS as a proportion of the building's replacement time (ReT). These ratios were derived numerically by Aljawhari et al. [31] and are listed in Table 4. T_0 is the time needed for activities like mobilisation, bidding, financing and permitting. An arbitrary T_0 of 100 days is assumed [44] due to the absence of standardized figures for Italy, but the actual value might be longer, depending on the local bureaucracy [45].

It is acknowledged that Equation (5) has limitations. It assumes that the structure is only re-occupied upon full recovery, which might not always be the case. It also omits other potential sources of indirect loss such as the loss of rental income [46] and interdependencies with other damaged buildings. Nevertheless, Equation (5) remains reasonable for shedding light on the role of indirect losses in life-cycle cost analyses, particularly when comparing different retrofit options/decisions in relative terms. The expected annual indirect loss (EAL_{ind}) is finally calculated by combining the LR_{ind} function with seismic hazard data and integrating the resulting curve over IM, based on Equation (6).

$$LR_{ind}(IM) = \frac{C_{rel}N_p}{ReC} \sum_{i=3}^{n} P(DS = DSi|IM)\,(DTRi \cdot ReT + T_0) \tag{5}$$

$$EAL_{ind} = \int_0^{\infty} LR_{ind}(\lambda)\, d\lambda \tag{6}$$

2.8 | Costs of Retrofit Interventions

Identifying optimal retrofits within budget constraints and life-cycle cost aspects necessitates calculating the cost of retrofit intervention ($C_{retrofit}$). $C_{retrofit}$ is expressed here as a function of the retrofit's geometry and layout, providing practical, rule-of-thumb estimates for both practitioners and future research. Deriving $C_{retrofit}$ begins with disaggregating construction works into individual activities and determining their unit prices, which are detailed in Table 5, based on the 2023 Price List of Civil Construction Works in Abruzzo region [47], where L'Aquila is located. The unit prices represent average values, covering the costs of materials, labour, and safety procedures but exclude technical costs like structural design and procurement. Only

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TABLE 4 | Adopted DTRs for different DSs and design-code levels.

Structure/DS	DS1 DTR1 [%]	DS2 DTR2 [%]	DS3 DTR3 [%]	DS4 DTR4 [%]
5-storey pre-code	9.2	23.1	49.6	79.6
5-storey low-code	9.5	24.3	53.0	82.9
5-storey high-code	9.9	25.6	58.8	84.3

TABLE 5 | Construction activities associated with steel and RC jacketing, and their unit prices.

Category	Activity	Unit price [€]	Steel jacketing	RC jacketing
Infill walls next to the column	Scrapping the paint	2.42/m^2	X	X
	Removing the plaster (up to 5 cm)	8.38/m^2	X	X
	Demolishing a single-leaf infill	7.44/m^2	X	X
	Re-building double-leaf infills via hollow bricks (external leaf is 12 × 25 × 25 cm and internal leaf is 8 × 25 × 25 cm)	76.93/m^2	X	X
	Adding mineral glass wool with a 30 mm thickness for thermal insulation	12.82/m^2	X	X
	Plastering using a cement mortar applied in three layers at each side	17.63/m^2	X	X
	Painting surfaces using a three-layer water-based paint	7.33/m^2	X	X
Slab part around the column ends	Scrapping the paint	2.42/m^2		X
	Removing the plaster (up to 5 cm)	8.38/m^2		X
	Removing floor tiles	15.28/ m^2		X
	Removing the screed	129.56/m^3		X
	Slab plastering work	19.37/m^2		X
	Painting the slab	7.33/m^2		X
	Adding a screed layer	15.44/m^2		X
	Laying floor tiles	48.53/m^2		X
Retrofit operations	Scrapping the paint layers off the existing column	2.42/m^2	X	X
	Removing the plaster from the existing column (up to 5 cm)	8.38/m^2	X	X
	Removing the concrete cover of the column	136.36/m^3		X
	Surface preparation prior to installing the jacket	62.44/m^2	X	
	Install a pre-fabricated elliptical S-235 steel jackets	5.01/kg	X	
	Fill gap between the jacket and column with cement grout	881.5/m^3	X	
	Paint the steel jacket	7.33/m^2	X	
	Erecting formwork around the column	37.18/m^2		X
	Add the steel cage (longitudinal rebar, hoops, dowels)	2260/ton		X
	Create holes in the existing column to put the dowels and add epoxy resin	111.63/kg		X
	Casting in-situ concrete mix	173.78/m^3		X
	Plastering the jacketed column	17.63/m^2		X
	Painting the jacketed column using a three-layer water-based paint	7.33/m^2		X
Disposal	Transportation costs to landfill	27.23/m^3	X	X
	Cost of landfilling	15.18/ton	X	X

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steel and RC jacketing are reported in Table 5, as they could be separated into individual activities. On the other hand, the costs of FRP wrapping exist in the Price List as a rough estimate that does not consider the number of retrofitted columns, FRP layers, or floor height. Hence, the FRP wrapping costs are drawn from Del Vecchio et al. [48], who compiled empirical data on retrofitted structures in L'Aquila.

Next, material quantities for column retrofit are estimated by assuming the geometric parameters as uniformly distributed random variables: beam widths and depths range from 0.25 to 0.40 m and 0.35 and 0.6 m, respectively. The as-built column's depth (h_c) and width (b_c) are between 0.3 and 0.6 m, while floor heights vary between 3.0 and 4.5 m [38, 49]. These parameters are considered uncorrelated [50]. 10,000 samples of geometric combinations are then generated using Monte Carlo simulation, and the C_{retrofit} value for each sample is computed by multiplying the unit prices in Table 5 with the respective material quantity.

For steel jacketing, C_{retrofit} is dominated by the jacket's thickness (t_j) and height (h_j). Therefore, a best-fit line is derived via least-square regression to model the relationship between t_j and cost realisations, as shown in Equation (7). The unit of t_j is mm, whilst h_j is in m. n_{ext} and n_{int} are the number of retrofitted external and internal columns, respectively. The regression achieves a coefficient of determination (R^2) of 0.88, with a root mean square error for the external (RMSE_{ext}) and internal (RMSE_{int}) columns equal to €$75h_j$.

$$\text{Steel Jacketing}: \; C_{\text{retrofit}}[€] = \sum_{i=1}^{n_{ext}} \left[387.2 + 77.3\, t_j\right] h_j + \sum_{i=1}^{n_{int}} \left[433.5 + 77.3\, t_j\right] h_j \quad (7)$$

Equation (8) shows the expected C_{retrofit} for FRP wrapping, adapted from Del Vecchio et al. [48], who estimated the costs of both retrofit and repair of seismic damage. The repair part is removed from Equation (8) as the as-built structure is assumed to be intact. The estimates of Del Vecchio et al. [48] were based on 2021 prices, so they are discounted by the ratio between the generic cost of FRP in the 2023 and 2021 editions of Abruzzo's Price List. The value of C_{retrofit} in Equation (8) comprises two parts: (1) the cost of wrapping FRP around columns defined in terms of the number of retrofitted columns (n_{col}), number of layers (n_{layer}), and h_j; (2) the cost of retrofitting external joints, which only requires their number ($n_{ext,joint}$). It is worth noting that retrofitting external joints with FRP can also be coupled with steel jacketing if the retrofit strategy requires both interventions.

$$\text{FRP Wrapping}: \; C_{\text{retrofit}}[€] = \sum_{i=1}^{n_{col}} \left[490.1 + 54.0\left(n_{layer} - 1\right)\right] h_j + \sum_{i=1}^{n_{ext,joint}} 1003.8 \quad (8)$$

For RC jacketing, C_{retrofit} is more complex due to its dependency on various reinforcement and geometric details. To simplify, only the most influential parameters are considered: t_j, h_c, b_c, and jacket's longitudinal rebar ratio (ρ_j). Monte Carlo sampling is utilised to simulate 10,000 combinations of jacket geometry and reinforcement, treating the previous parameters as uniformly distributed random variables: ρ_j is between 0.3% and 1.5%, and t_j is within 0.05 and 0.20 m, while both h_c and b_c range from 0.3 to 0.6 m. A least-square multiple regression line is now fitted to represent the relationship between these variables and the corresponding C_{retrofit}, as per Equation (9). R^2 is 0.92, with RMSE_{ext} of €$18h_j$ and RMSE_{int} of €$22h_j$. The units are metre for t_j, h_c, b_c, and h_j, whereas ρ_j is plugged in as a percentage (%).

$$\text{RC Jacketing}: \; C_{\text{retrofit}}[€] = \sum_{i=1}^{n_{ext}} [433.7b_c + 455.1h_c + 927.5t_j + 101.6\rho_j + 261.0]h_j + \sum_{i=1}^{n_{int}} [456.1b_c + 454.8h_c + 927.5t_j + 101.6\rho_j + 433.3]h_j \quad (9)$$

Table 6 summarises the total C_{retrofit} for all retrofit configurations and strategies, determined via Equations (7)–(9). Figure 7 depicts the relationship between C_{retrofit} (as % of ReC) and the improvement in seismic performance in terms of CDR_{LS}. The ReC is €984/m^2 [51], discounted by the ratio of average construction prices in 2023 and 2014, as indicated in Abruzzo's Price List. For FRP wrapping and steel jacketing, Figure 7a,b reveals a sharp increase in C_{retrofit} at lower CDR_{LS} levels (<65%), followed by a moderate rise at the remaining CDR_{LS} levels. In contrast, Figure 7c shows that C_{retrofit} for RC jacketing rises steeply at all CDR_{LS} values. This trend reflects the nature of incremental retrofit using RC jacketing, where marginal costs escalate rapidly as jackets are expanded to additional columns, leading eventually to retrofitting of the entire structure. For FRP wrapping and steel jacketing, the incremental process reaches a plateau once the number of retrofitted columns stabilises. At this stage, only minor adjustments to parameters like t_j and n_{layer}, are needed, which have modest impact on C_{retrofit}. Depending on CDR_{LS}, C_{retrofit} for FRP wrapping and steel jacketing ranges from as low as 3% to as high as 11% of the building's ReC. The RC jacketing is slightly more expensive at CDR_{LS} below 70%, but sees a substantial increase beyond this threshold, reaching up to 18% of ReC.

2.9 | Embodied Carbon of Retrofit Interventions

The current study's approach to life-cycle sustainability covers not only the carbon induced by earthquake damage, but also that generated during the implementation of retrofit interventions ($\text{CO}_{2,\text{retrofit}}$). To quantify $\text{CO}_{2,\text{retrofit}}$, carbon factors are adopted, which estimate the greenhouse gas emissions associated with the production of construction materials, expressed in kgs of CO_2e per unit quantities (e.g., tons, m^3) [36]. The carbon footprint for any construction activity can be easily obtained by multiplying the required material quantity with the appropriate factor. The factors used here (see Table 7) are obtained from Aljawhari et al. [38], who compiled empirical data from more than 230 Italian products. These factors account solely for the emissions of material production as they are the largest contributors. Therefore, the emissions of demolition, waste processing and disposal are added separately. The carbon of demolition is estimated at 3.4 kgCO_2e per 1 m^2 of building area, while that related to waste

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TABLE 6 | Costs of retrofit interventions for the three strategies with their different configurations and CDR_{LS}.

FRP wrapping		Steel jacketing		RC jacketing	
CDR [%]	$C_{retrofit}$ **[€]**	**CDR [%]**	$C_{retrofit}$ **[€]**	**CDR [%]**	$C_{retrofit}$ **[€]**
42 (as-built)	0	42 (as-built)	0	42 (as-built)	0
54	89,053	55	90,517	45	29,362
62	100,055	61	102,891	65	108,374
65	111,058	80	115,265	78	146,753
73	114,431	83	116,135	93	163,631
76	115,645	86	117,006	105	178,312
80	136,828	89	118,553	109	178,531
86	140,201	94	139,074	114	198,819
92	143,574	97	139,944	126	199,754
94	145,733	106	143,909	130	231,220
98	149,105	115	148,744	135	259,008
105	151,264	121	152,032	139	259,394
109	163,481	126	153,579	143	260,382
110	164,695	132	155,320	147	260,675
N.A.	N.A.	138	158,414	153	261,291
N.A.	N.A.	147	159,284	156	261,584
N.A.	N.A.	N.A.	N.A.	167	267,544
N.A.	N.A.	N.A.	N.A.	181	270,667

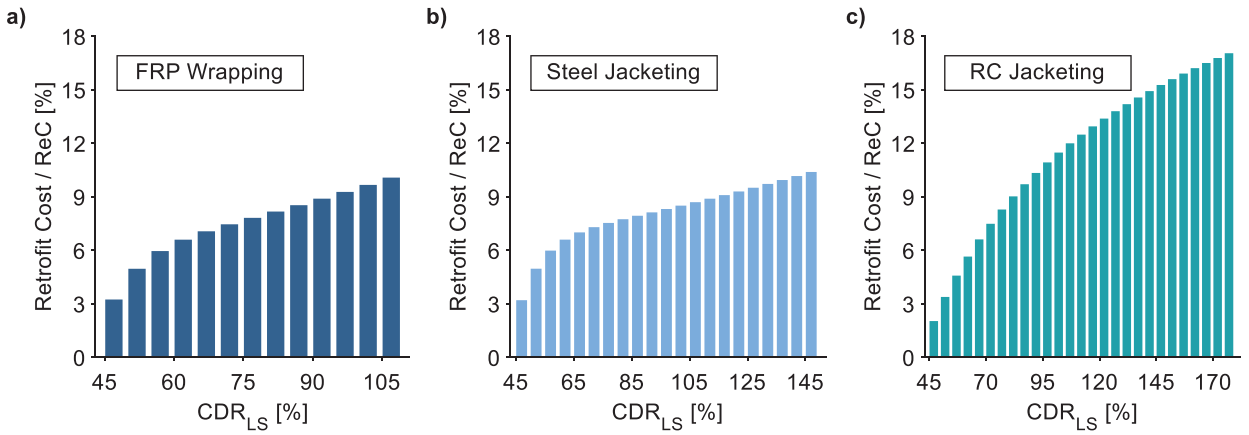


FIGURE 7 | Cost of retrofit intervention normalised by ReC for: (a) FRP wrapping; (b) steel jacketing; and (c) RC jacketing.

processing/disposal is quantified at a rate of 0.013 $kgCO_2e$ per 1 kg of waste [37]. It is assumed that 90% of waste materials are recycled, while the remaining 10% are landfilled [52].

To obtain $CO_{2,retrofit}$, it is essential to identify construction activities that result in non-negligible emissions (see Table 8) and then estimate the necessary material quantities. Activities with limited carbon impact, like technical efforts and installation of scaffolding/shoring, are excluded from Table 8. The assumptions of geometric variability for the retrofitted columns and fitting procedures align with those used for retrofit costs, $C_{retrofit}$ (see Section 2.8). The variability of carbon factors in Table 7 is further considered by generating 10,000 samples via Monte Carlo simulation. The total $CO_{2,retrofit}$ values are finally computed by multiplying the quantities of construction materials with the corresponding samples of carbon factors, then adding the contribution of demolition and waste disposal.

The expected $CO_{2,retrofit}$ in $kgCO_2e$ for steel jacketing is expressed by a least-square regression line to relate carbon emissions to t_j in mm and h_j in m, as per Equation (10). The same applies to the FRP wrapping, where the independent variable becomes n_{layer}, as shown in Equation (11). The R^2 for both expressions exceed 0.85, with $RMSE_{ext}$ and $RMSE_{int}$ equal to $23h_j$ $kgCO_2e$ for the steel jacketing and 35_j $kgCO_2e$ for the FRP wrapping. The formulation of $CO_{2,retrofit}$ for RC jacketing is a function of t_j, h_c,

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TABLE 7 | Average carbon factors, in kgs of CO_2e, for different construction materials, after Aljawhari et al. [38].

Product/ material category	Average carbon factor [kgCO_2e]	Standard deviation (σ)	Unit quantity
Concrete	299.00	113.000	m^3
Steel reinforcement	729.00	94.000	ton
Sand (fines)	31.25	9.510	ton
Cement	733.75	132.130	ton
Clay bricks	0.24	0.056	kg
Epoxy resin	6.64	1.160	kg
FRP sheets	5.00	0.00	kg
Hot-rolled steel profiles	1030.05	2.9	ton
Insulation for external walls	3.37	1.040	m^2
Wall paint	6.21	1.600	kg
Flooring tiles	15.90	6.044	m^2

b_c (all in m), and ρ_j(in %). A least-square multiple regression fit yields Equation (10), with R^2 of 0.84 and $RMSE_{ext}$ and $RMSE_{int}$ equal to $23h_j$ kgCO_2e.

$$\text{Steel Jacketing}: \ CO_{2,\text{retrofit}}\left[\text{kgCO}_2\text{e}\right] = \sum_{i=1}^{n_{ext}}\left[181.6 + 15.9\,t_j\right]h_j + \sum_{i=1}^{n_{int}}\left[223.3 + 15.9\,t_j\right]h_j \tag{10}$$

$$\text{FRP Wrapping}: \ CO_{2,\text{retrofit}}\left[\text{kgCO}_2\text{e}\right] = \sum_{i=1}^{n_{ext}}\left[233.4 + 28.4\,n_{layer}\right]h_j + \sum_{i=1}^{n_{int}}\left[288.4 + 28.4\,n_{layer}\right]h_j + \sum_{i=1}^{n_{ext,joint}} 201.8 \tag{11}$$

$$\text{RC Jacketing}: \ CO_{2,\text{retrofit}}\left[\text{kgCO}_2\text{e}\right] = \sum_{i=1}^{n_{ext}}\left[146.6b_c + 158.5h_c + 695.0t_j + 35.8\rho_j + 188.5\right]h_j + \sum_{i=1}^{n_{int}}\left[157.3b_c + 158.5h_c + 695.0t_j + 35.8\rho_j + 295.0\right]h_j \tag{12}$$

Applying Equations (10)–(12), the total $CO_{2,\text{retrofit}}$ for all retrofit configurations and strategies is evaluated and reported in Table 9. Figure 8 depicts the $CO_{2,\text{retrofit}}$ normalised by the building's $ReCO_2$, assumed to be 385 kgCO_2e per 1 m^2 of construction area [38, 53]. The trends in $CO_{2,\text{retrofit}}$ resemble those observed in cost analysis, particularly regarding the sharpness and marginal embodied carbon (see Section 2.8). Interestingly, the normalised $CO_{2,\text{retrofit}}$ for FRP wrapping and steel jacketing in Figure 8a,b tends to be lower than the respective normalised C_{retrofit}. In contrast, the normalised $CO_{2,\text{retrofit}}$ of RC jacketing is significantly larger than its normalised C_{retrofit}, reaching nearly 25% at high CDR_{LS} levels, according to Figure 8c. This disparity may have crucial implications on sustainability-focused decision-making, as discussed later.

2.10 | Life-Cycle Analysis

Optimising retrofit designs necessitates analysing a building's life-cycle cost as outlined by Equation (13). This analysis includes both the initial investment required for retrofitting, C_{retrofit}, and the repair costs incurred due to seismic damage over the building's service life (T). The latter is represented as the expected present value of future repair costs, which takes into consideration a discount rate (r) and the remaining T [54]. In our analysis, T is set at 50 years and r at 3.0% [55]. The life-cycle cost can be further refined to account for the effect of indirect losses by adding EAL_{ind}, as per Equation (14).

$$\text{Life cycle cost (without indirect loss)} = C_{\text{retrofit}} + \frac{\left(1 - e^{-rT}\right)}{r}\text{EAL} \cdot \text{ReC} \tag{13}$$

$$\text{Life cycle cost (with indirect loss)} = C_{\text{retrofit}} + \frac{\left(1 - e^{-rT}\right)}{r}\left(\text{EAL} + \text{EAL}_{\text{ind}}\right) \cdot \text{ReC} \tag{14}$$

In terms of sustainability, this study explores two distinct approaches for handling the carbon generated by repairing seismic damage. The first consists of evaluating life-cycle carbon as per Equation (15), which accounts for contributions from both the retrofit intervention, $CO_{2,\text{retrofit}}$, and seismic damage over the building's T. The second approach is monetising emissions through a 'carbon tax' ($CO_{2,\text{tax}}$) [56]. This tax assigns a price on the generated emissions by setting a specific rate per ton of CO_2eq, allowing carbon to be explicitly factored in conventional life-cycle costs, as shown by Equation (16).

$$\text{Life cycle carbon} = CO_{2,\text{retrofit}} + \text{T} \cdot \text{EAC} \cdot \text{ReCO}_2 \tag{15}$$

$$\text{Life cycle cost} = C_{\text{retrofit}} + \frac{\left(1 - e^{-rT}\right)}{r}\left(\text{EAL} + \text{EAL}_{\text{ind}}\right)\text{ReC} + \left(CO_{2,\text{retrofit}} + \frac{\left(1 - e^{-rT}\right)}{r}\text{EAC} \cdot \text{ReCO}_2\right)CO_{2,\text{tax}} \tag{16}$$

It is acknowledged that life-cycle calculations in Equations (13)–(16) do not address contributions like use/operation and ordinary maintenance. This assumption is deemed reasonable, though, as such expenses are relatively similar across both the as-built and retrofitted configurations, so they are unlikely to skew the comparison process. An exception to this might occur if the retrofit strategy involves major energy upgrades (e.g., thermal insulation) that remarkably alter the life-cycle carbon footprint and overall costs [20].

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TABLE 8 | Construction activities generating carbon emissions for the three retrofit strategies.

Category	Activity	FRP wrapping	Steel jacketing	RC jacketing
Infill walls next to the column	Re-building double-leaf infills via hollow bricks (external leaf is 12 × 25 × 25 cm and internal leaf is 8 × 25 × 25 cm)	X	X	X
	Adding mineral glass wool with a 30 mm thickness for thermal insulation	X	X	X
	Plastering using a cement mortar applied in three layers at each side	X	X	X
	Painting surfaces using a three-layer water-based paint	X	X	X
Slab part around the column ends	Slab plastering work			X
	Painting the slab			X
	Adding a screed layer			X
	Laying floor tiles			X
Retrofit operations	Add a layer of epoxy adhesive on the existing column for each FRP layer	X		
	Wrap the required amount of FRP layers along the entire column height	X		
	Install the pre-fabricated elliptical S-235 steel jackets		X	
	Fill the gap between the jacket and column with cement grout		X	
	Paint the steel jacket		X	
	Add the steel cage (longitudinal rebar, hoops, dowels)			X
	Create holes in the existing column to put the dowels and add epoxy resin			X
	Casting in situ concrete mix			X
	Plastering the jacketed column	X		X
	Painting the jacketed column using a three-layer water-based paint	X		X
Disposal	Landfill and waste processing	X	X	X

TABLE 9 | Carbon emissions of retrofit interventions for the three strategies with their different configurations and CDR_{LS}.

FRP wrapping		Steel jacketing		RC jacketing	
CDR [%]	$CO_{2,retrofit}$ [$kgCO_2e$]	CDR [%]	$CO_{2,retrofit}$ [$kgCO_2e$]	CDR [%]	$CO_{2,retrofit}$ [$kgCO_2e$]
42 (as-built)	0	42 (as-built)	0	42 (as-built)	0
54	31,413	55	28,623	45	15,632
62	37,905	61	34,185	65	58,265
65	44,397	80	39,747	78	78,950
73	46,175	83	39,926	93	88,818
76	46,815	86	40,105	105	96,634
80	54,876	89	40,423	109	96,722
86	56,654	94	46,645	114	108,090
92	58,432	97	46,823	126	108,426
94	59,570	106	47,638	130	124,812
98	61,348	115	48,632	135	138,867
105	62,486	121	49,308	139	139,017
109	69,618	126	49,626	143	139,833
110	70,258	132	49,984	147	139,939
N.A.	N.A.	138	50,620	153	140,159
N.A.	N.A.	147	50,799	156	140,265
N.A.	N.A.	N.A.	N.A.	167	142,402
N.A.	N.A.	N.A.	N.A.	181	144,349

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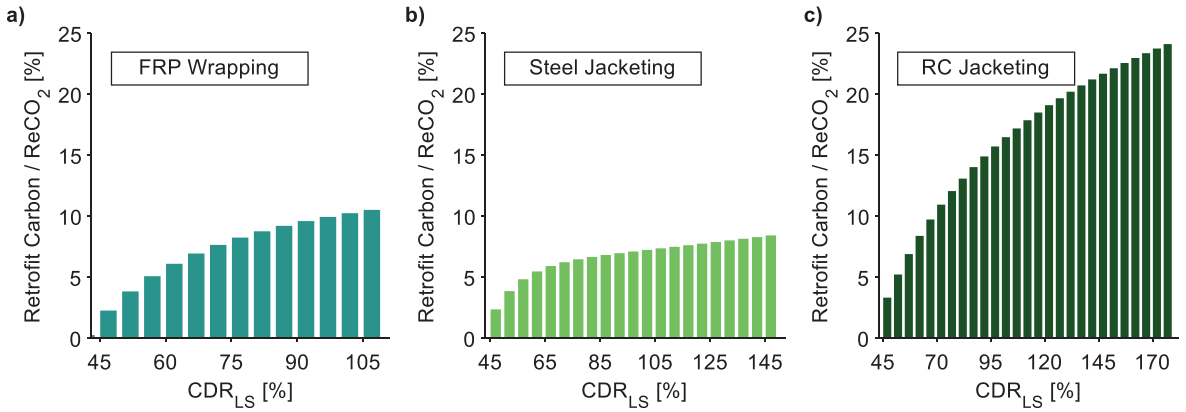


FIGURE 8 | Carbon of retrofit intervention normalised by $ReCO_2$ for (a) FRP wrapping; (b) steel jacketing; and (c) RC jacketing.

TABLE 10 | EAL, EAL_{ind}, and EAC for the three retrofit strategies with their different configurations and CDR_{LS}.

FRP wrapping				Steel jacketing				RC jacketing			
CDR [%]	EAL [%]	EAL_{ind} [%]	EAC [%]	CDR [%]	EAL [%]	EAL_{ind} [%]	EAC [%]	CDR [%]	EAL [%]	EAL_{ind} [%]	EAC [%]
42	1.423	0.603	1.330	42	1.423	0.603	1.330	42	1.423	0.603	1.330
54	1.185	0.270	1.093	55	1.284	0.296	1.148	45	1.394	0.474	1.309
62	1.091	0.203	1.002	61	1.167	0.229	1.040	65	1.119	0.250	1.130
65	1.056	0.171	0.973	80	1.101	0.100	0.986	78	1.041	0.155	1.030
73	1.002	0.137	0.924	83	1.076	0.096	0.967	93	0.685	0.068	0.694
76	0.979	0.124	0.906	86	1.039	0.085	0.860	105	0.591	0.055	0.582
80	0.926	0.106	0.854	89	0.998	0.076	0.828	109	0.571	0.047	0.561
86	0.903	0.087	0.836	94	0.981	0.066	0.818	114	0.545	0.037	0.544
92	0.848	0.070	0.781	97	0.945	0.059	0.858	126	0.534	0.031	0.531
94	0.833	0.067	0.769	106	0.934	0.046	0.852	130	0.514	0.033	0.502
98	0.869	0.063	0.802	115	0.909	0.032	0.832	135	0.508	0.030	0.494
105	0.849	0.050	0.791	121	0.891	0.030	0.813	139	0.475	0.028	0.455
109	0.813	0.043	0.766	126	0.895	0.026	0.817	143	0.475	0.026	0.458
110	0.798	0.042	0.771	132	0.970	0.023	0.806	147	0.463	0.024	0.447
N.A.	N.A.	N.A.	N.A.	138	0.949	0.020	0.781	153	0.453	0.022	0.436
N.A.	N.A.	N.A.	N.A.	147	0.936	0.016	0.801	156	0.443	0.021	0.427
N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	167	0.417	0.018	0.394
N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	181	0.410	0.014	0.387

3 | Results and Discussion

3.1 | Earthquake-Damage Consequences

Prior to conducting life-cycle analyses as per Section 2.10, earthquake consequences, expressed in terms of EAL for repair costs, EAC for carbon emissions, and EAL_{ind} for indirect losses, must be evaluated for the as-built and retrofitted configurations. Table 10 reports the resulting values computed via Equations (2), (4) and (6), showing that both EAL and EAC are heavily influenced by low damage levels (DS1 and DS2) as they occur at smaller IMs with high λ values. It is observed that the reduction of EAL due to retrofit is consistent among the three retrofit strategies for CDR less than 90%. Beyond this threshold, RC jacketing becomes the most effective strategy due to its superior improvement of DS1 and DS2 fragility relationships, and its ability to transform the plastic mechanism from(local) column-sway to (global) beam-sway. The reduction in EAC has a similar trend across the three strategies, but the EAC are generally lower than the EAL values, primarily because the DCR for DS1 is smaller than its DLR counterpart. The difference between the two metrics, however, remains under 10%. Interestingly, the three strategies are consistent in reducing EAL_{ind}, irrespective of the CDR_{LS} level. This is because indirect losses are associated with the building's response under high levels of non-linear deformation (i.e., DS3-DS4), which can be enhanced effectively by the three retrofit strategies.

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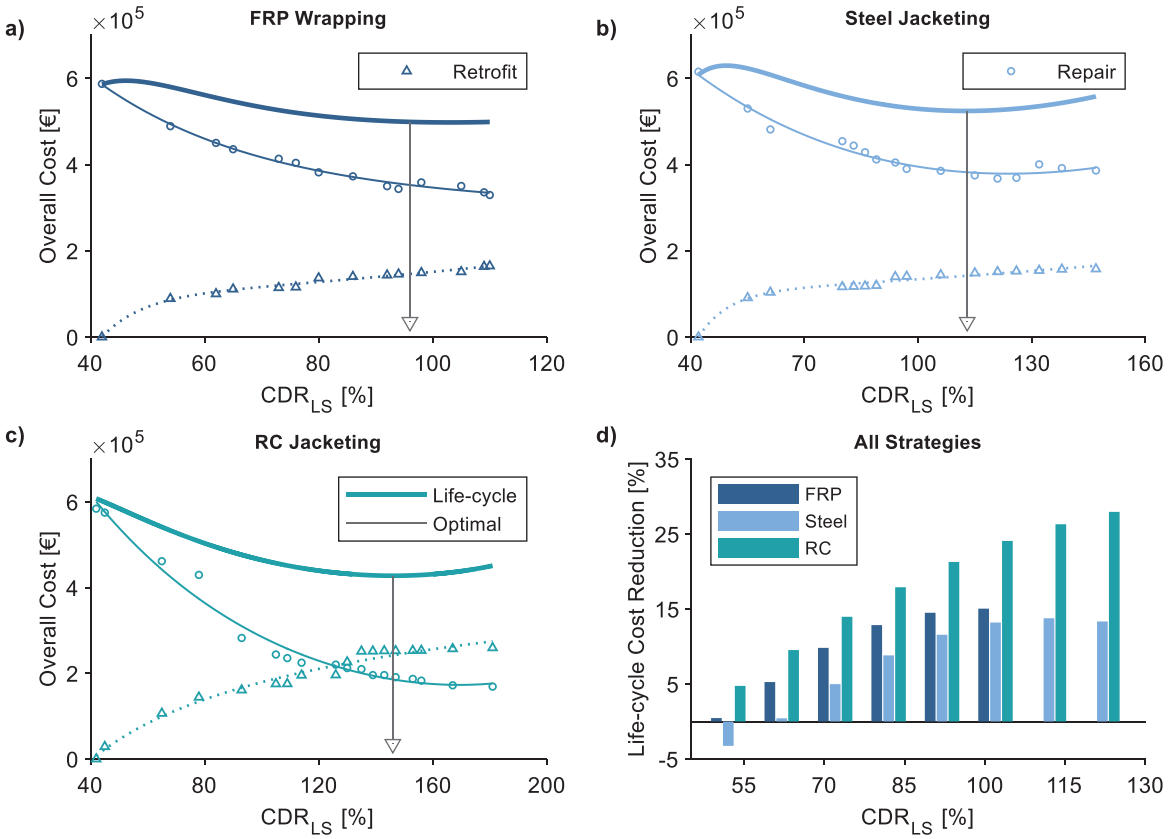


FIGURE 9 | (a–c) Life-cycle costs including seismic damage repair and retrofit intervention versus CDR_{LS} for the FRP wrapping, steel jacketing, and RC jacketing; (d) % reduction in life-cycle costs due to retrofit relative to the as-built.

3.2 | Life-Cycle Cost Optimisation

Life-cycle costs are first analysed, accounting for both $C_{retrofit}$ and repair costs of seismic damage, as per Equation (13). Figure 9a-c shows the variation of these life-cycle costs as a function of CDR_{LS} levels achieved through incremental retrofit. The corresponding trends of $C_{retrofit}$ and repair costs of seismic damage are also displayed. It is observed that incremental retrofit results in a steady decrease in life-cycle costs till reaching an optimal point (or CDR_{LS}) highlighted by the arrow. Beyond this point, further interventions lead to higher life-cycle costs because the reduction in repair costs no longer offsets the rising $C_{retrofit}$. The optimal CDR_{LS} varies significantly by retrofit strategy: For FRP wrapping, it is 95%, slightly below the Eurocode 8–3 requirement of 100% [57]. In contrast, the optimal CDR_{LS} values for steel and RC jacketing are remarkably larger, at 117% and 146%, respectively. This suggests that satisfying code-based performance objectives may not always produce an optimal retrofit solution as this heavily depends on the retrofit strategy. This is particularly evident for RC jacketing, which delivers a substantial life-cycle cost reduction at high CDR_{LS}.

Figure 9d compares the percentage of reduction in life-cycle cost achieved by retrofitted configurations relative to the as-built structure. The results clearly indicate that RC jacketing offers the greatest reduction of life-cycle costs, outperforming both FRP wrapping and steel jacketing at all CDR_{LS} levels, despite being the most expensive. Therefore, RC jacketing proves to be the most favourable strategy when life-cycle costs ($C_{retrofit}$ plus repair of seismic damage) are the primary decision-making criterion, assuming no budget constraints.

3.3 | Indirect Loss Considerations in Life-Cycle Cost Optimisation

Indirect losses can surprisingly impact decisions on seismic retrofit, especially at lower CDR_{LS} levels. To clarify that, Figure 10 shows the indirect loss reduction, for each retrofitted configuration, relative to the as-built structure across various CDR_{LS} levels. Unlike direct losses (repair costs), FRP wrapping and steel jacketing outperform RC jacketing at $CDR_{LS} < 80\%$ because indirect losses rely on the building's performance under high DSs (i.e., DS3-DS4), as per Equation (5). To withstand DS3-DS4, the as-built structure needs a significant improvement in deformation ductility, which is more effectively attained by FRP wrapping and steel jacketing of columns as they provide continuous lateral confinement.

The effect of indirect loss on life-cycle cost is now analysed using Equation (14). The updated life-cycle costs for the three retrofit strategies are illustrated in Figure 11a-c as a function of CDR_{LS}, along with the optimal CDR_{LS} and breakdown of life-cycle cost components ($C_{retrofit}$, repair costs, and indirect losses). The optimal CDR_{LS} values are almost identical to those in Figure 9a-c, suggesting that a life-cycle cost that includes $C_{retrofit}$ and repair costs alone may be sufficient for determining the optimal retrofit configuration.

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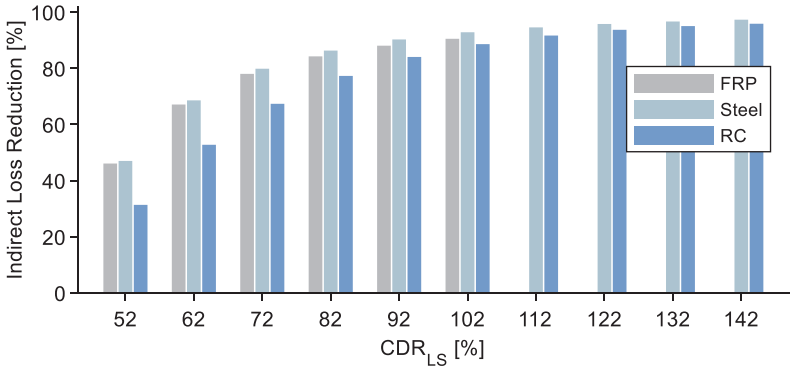


FIGURE 10 | CDR_{LS} versus the reduction in indirect loss due to the three retrofit strategies.

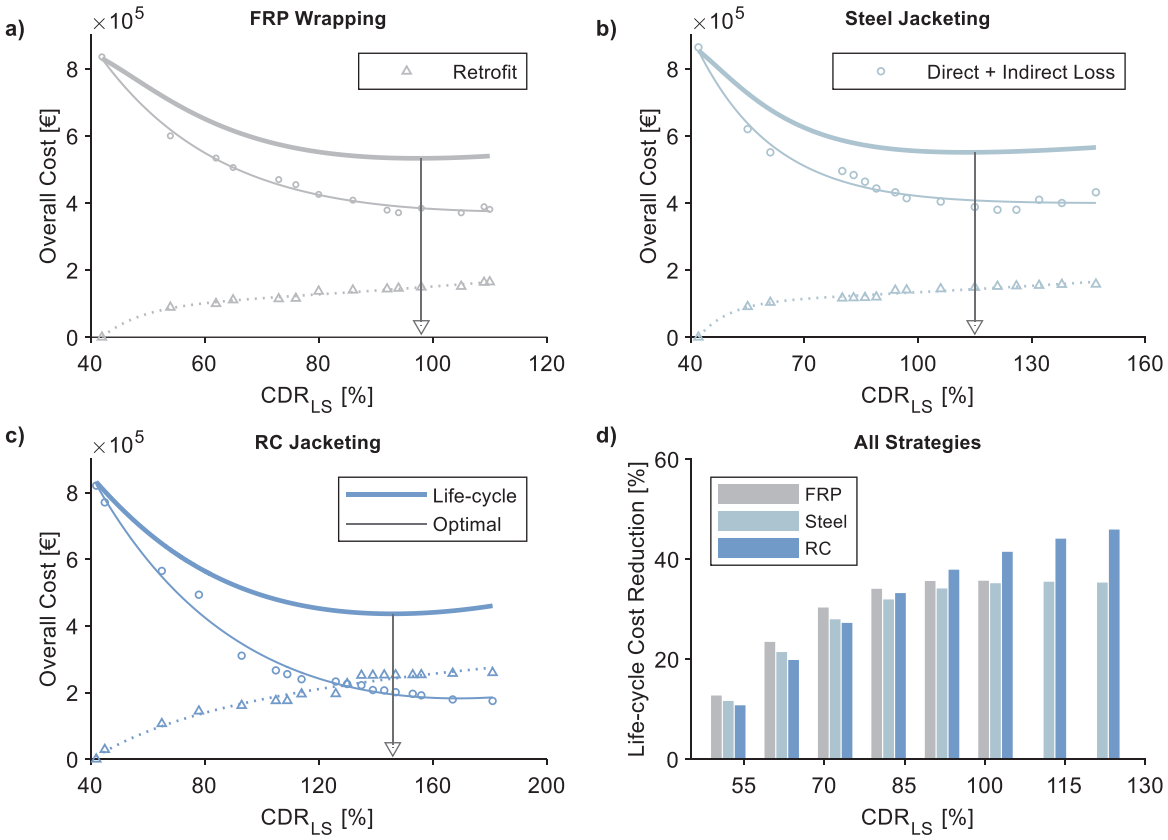


FIGURE 11 | (a–c) life-cycle costs including indirect losses, seismic damage repair and retrofit intervention versus CDR_{LS} for the FRP wrapping, steel jacketing, and RC jacketing; (d) % reduction in life-cycle costs due to retrofit relative to the as-built.

Figure 11d presents the reduction percentage in life-cycle cost for the retrofitted configurations relative to the as-built structure for all CDR_{LS} levels. Unlike Figure 9d, which positions RC jacketing as the superior strategy overall, Figure 11d unveils that FRP wrapping and steel jacketing yield greater reductions in life-cycle costs for CDR_{LS} levels less than 80%, given their better control of indirect losses (see Figure 10). However, RC jacketing becomes the best option at $CDR_{LS} > 90\%$ as it shifts the building's lateral-sway mechanism to a global beam-sway one, leading to a significant decline in repair costs that outweighs the benefits of minimising indirect losses.

Moreover, indirect losses can alter retrofit decisions in regions with lower seismic hazard. Figure 12a displays the variation of life-cycle costs (excluding indirect losses) as a function of CDR_{LS} for FRP wrapping, assuming the building is in the low-hazard zone (i.e., Florence in Figure 5). Conversely, Figure 12b incorporates indirect losses as part of life-cycle costs. Without considering indirect losses, Figure 12a indicates that FRP wrapping does not reduce life-cycle costs, making the 'no-intervention' option the most viable. When indirect losses are factored in, as per Figure 12b, FRP wrapping provides a significant life-cycle cost reduction. These findings underscore the role of indirect-loss

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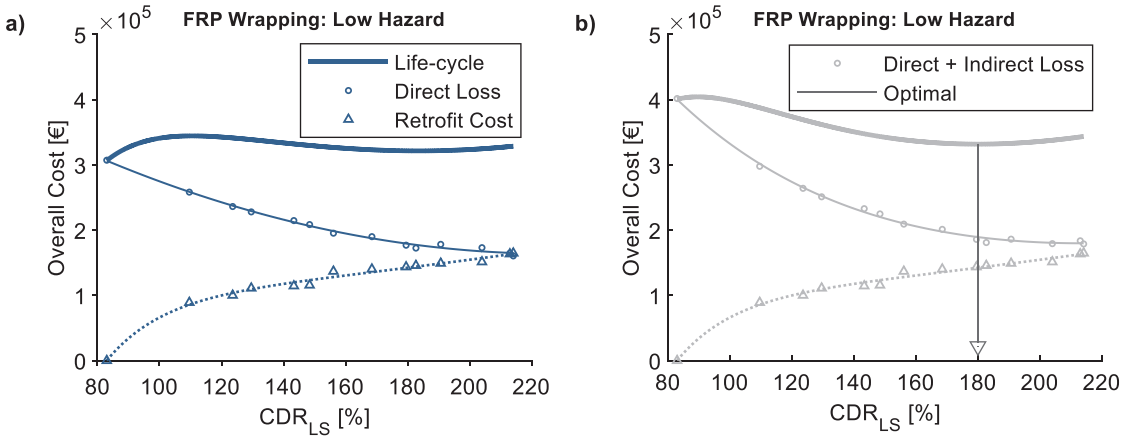


FIGURE 12 | Life-cycle costs of FRP wrapping at the low hazard level: (a) excluding indirect loss and (b) including indirect loss.

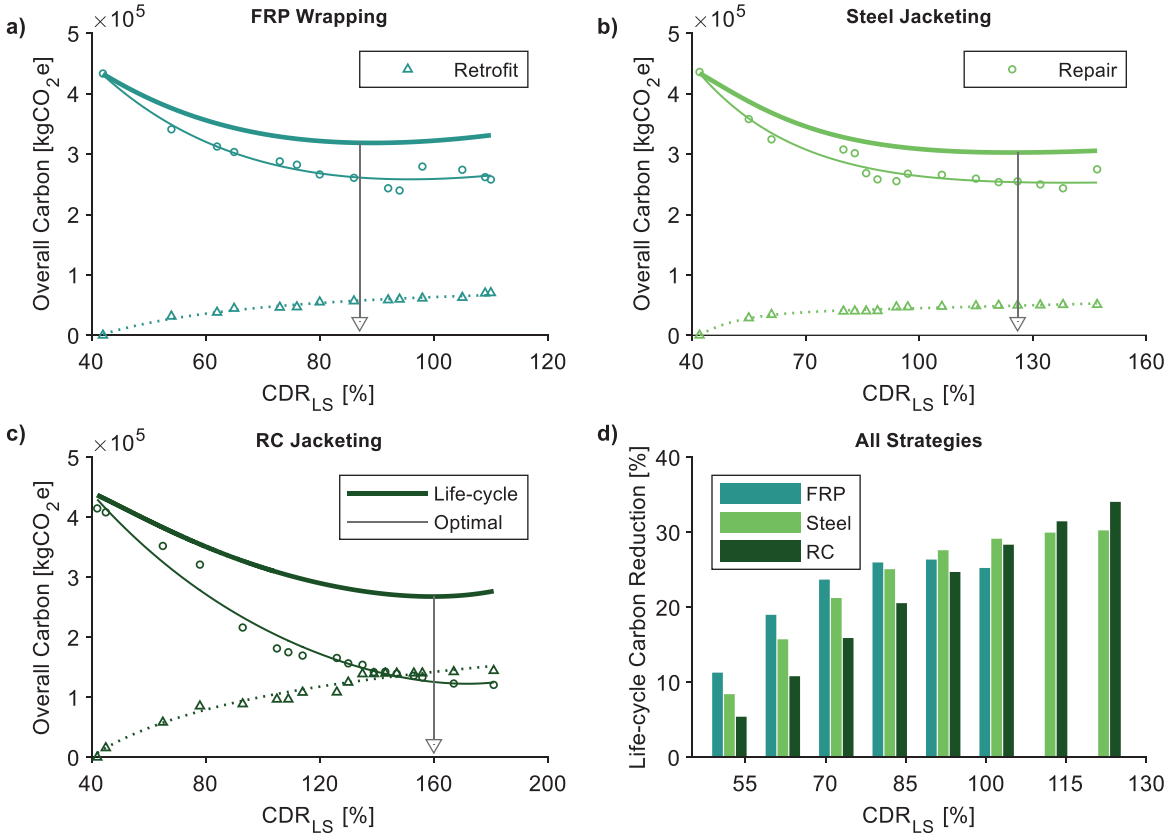


FIGURE 13 | (a–c) life-cycle carbon including seismic damage repair and retrofit intervention versus CDR_{LS} for the FRP wrapping, steel jacketing, and RC jacketing; (d) % reduction in life-cycle carbon due to retrofit relative to the as-built.

estimation to enable more meaningful comparisons between retrofit strategies and their feasibility.

3.4 | Sustainability Considerations in Life-Cycle Optimisation

Although earthquake environmental impacts (e.g., carbon emissions) have received increased attention in recent years, the value of integrating them into retrofit design and selection remains uncertain, primarily due to their strong correlation with repair costs (see Section 2.6). It is often argued that repair costs alone may suffice as a proxy for environmental impacts in decision-making. To investigate this, life-cycle carbon is computed via Equation (15) and plotted in Figure 13a-c against CDR_{LS} values for the three retrofit strategies. The plots also show the gradual increase of $CO_{2,retrofit}$ alongside the corresponding reduction of earthquake-induced carbon as the building undergoes incremental retrofitting. The optimal values CDR_{LS} minimising life-cycle carbon differs by less than 10% from those obtained for life-cycle costs (see Figure 9). This minor difference means that optimising retrofit designs for life-cycle costs can simultaneously optimise

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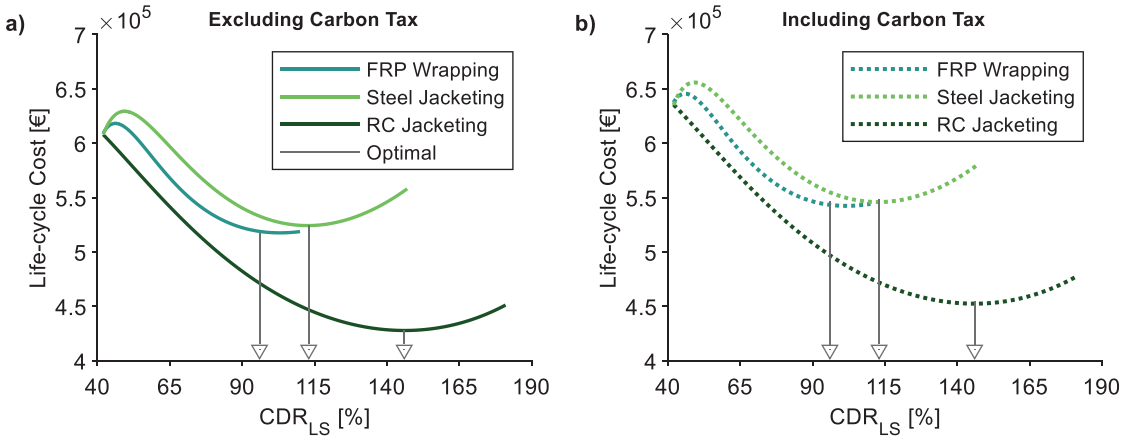


FIGURE 14 | Life-cycle costs versus CDR_{LS}: (a) excluding carbon tax; and (b) including carbon tax.

carbon emissions and vice versa. Figure 13a-c also indicates that the reduction in carbon emissions of seismic damage follows the same trend of repair cost reduction, confirming that the retrofit with the lowest repair costs is automatically the one with the least emissions due to seismic damage.

Unexpectedly, the life-cycle carbon does not follow a similar pattern as life-cycle costs, driven by the considerably lower $CO_{2,retrofit}$ for FRP wrapping and steel jacketing compared to RC jacketing. This demonstrates that the retrofit strategy with the least life-cycle cost is not necessarily the most sustainable (i.e., lowest life-cycle carbon), despite the strong correlation between repair costs and carbon resulting from seismic damage. This is evident in Figure 13d, which reports the reduction percentage in life-cycle carbon for the retrofitted configurations relative to the as-built structure. At CDR_{LS} levels <80%, FRP wrapping and steel jacketing outperform RC jacketing, offering reductions up to a 24% in life-cycle carbon, compared to the RC jacketing's 5%–14%. This constitutes a sharp diversion from the trends in Figure 9d, where RC jacketing consistently achieves the least life-cycle costs at all CDR_{LS} levels. As an example, at CDR_{LS} of 65%, opting for FRP wrapping over RC jacketing would be a far more sustainable option, even if it entails slightly higher life-cycle costs, reflecting the importance of addressing both economic and environmental criteria in decision-making.

Alternatively, it is possible to explicitly integrate sustainability with life-cycle costs by monetising carbon emissions via a carbon tax, $CO_{2,tax}$, according to Equation (16). Although $CO_{2,tax}$ is not yet implemented in Italy, many European countries have taxes already. According to the Carbon Pricing Dashboard [58], the 2023 $CO_{2,tax}$ within the European Economic Area ranges from €2.00 (Estonia) to €120.16 (Liechtenstein and Switzerland) per 1 ton of CO_2e, with an average of €44.50. Figure 14a shows the ordinary life-cycle costs as a function of CDR_{LS} for the three retrofit strategies, while Figure 14b plots the same data, incorporating the effects of $CO_{2,tax}$, as per Equation (16). The $CO_{2,tax}$ is set at €120.16 per 1 ton CO_2e to reflect the upper bound of European rates. The patterns in life-cycle costs in both scenarios are identical, with the $CO_{2,tax}$ introducing only a minor vertical shift in the curves without altering the general trend or optimal CDR_{LS} levels. This is because the $CO_{2,tax}$ reduces the order of magnitude of carbon estimates, making their overall impact on life-cycle costs minor (below 10%). Adopting a lower $CO_{2,tax}$ would render this effect even more negligible.

In summary, the role of carbon emissions in the design, selection, and comparison between different seismic retrofit strategies is contingent on the decision-maker's priorities regarding environmental considerations. If sustainability is a core objective of the decision criteria, then life-cycle carbon computed using Equation (15) can govern the selection of suitable retrofit strategies, depending on CDR_{LS}. On the contrary, if economic factors are the pure decision-making drivers, monetising carbon emissions through $CO_{2,tax}$ has a limited value, serving primarily to fine-tune life-cycle cost estimates without influencing design choices or optimisation of retrofit strategies.

4 | Practical Considerations for Retrofit Design Optimisation

4.1 | Code-Based Retrofit Design Versus Optimised Design

The preceding sections demonstrate the strong influence of various metrics on decision-making outcomes for seismic retrofit. To enable more meaningful comparisons and informed decisions, it is advisable to individually optimise the design of each retrofit strategy to achieve a desired target before comparing it to other options. These targets may include minimal life-cycle costs, a specific level of risk/consequence of interest, among others. To emphasise the importance of design optimisation, Table 11 compares the retrofit strategies under three scenarios: (1) meeting the Eurocode 8–3 requirement of CDR_{LS} equal to 100% (i.e., no optimisation); (2) optimising the retrofit design to achieve a 25% reduction in life-cycle costs; and (3) optimising the design to attain the lowest possible life-cycle costs.

Table 11 highlights the complexities of the first scenario (CDR_{LS} of 100%). While RC jacketing has the lowest life-cycle costs, for example, FRP wrapping is the most sustainable, and steel jacketing is the least expensive. In the second scenario, all three strategies have identical life-cycle costs. Both FRP wrapping and steel jacketing offer similar reductions in life-cycle carbon, but FRP wrapping is the most cost-effective, and, therefore, the

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TABLE 11 | Comparison between different retrofitting scenarios and performance objectives.

Parameter/objective	Scenario 1: Eurocode 8–3 performance			Scenario 2: 25% reduction in life-cycle cost (with indirect loss)			Scenario 3: Minimal life-cycle cost		
	FRP	Steel jacketing	RC jacketing	FRP	Steel jacketing	RC jacketing	FRP	Steel jacketing	RC jacketing
CDR_{LS} [%]	100	100	100	68	76	73	97	113	146
Retrofit cost [€]	151,200	134,100	179,800	113,000	119,000	122,400	147,500	142,100	241,900
Direct loss [€]	346,500	395,400	285,900	424,600	448,400	397,900	347,000	382,100	185,800
Indirect loss [€]	35,900	25,500	26,800	62,400	32,700	79,700	12,800	26,100	8,900
Life-cycle cost (no indirect losses) [€]	497,700	529,500	465,700	537,600	567,300	520,300	494,500	524,200	427,700
Life-cycle cost with indirect losses [€]	533,600	555,000	492,500	600,000	600,000	600,000	507,300	550,300	436,000
Life-cycle carbon [$kgCO_2e$]	308,500	322,000	318,000	335,700	335,000	365,800	320,400	303,700	270,800

preferred choice. The third scenario is more straightforward: RC jacketing, despite being the most expensive, achieves the lowest life-cycle costs and highest sustainability, making it the optimal choice in the absence of budgetary constraints. It should be noted that seismic codes and guidelines worldwide might specify seismic performance and hazard levels other than the life safety and the 475-year return period suggested by Eurocode 8-3, but they are outside the scope of this work [59, 60].

4.2 | Practical Optimisation for Retrofit Design

Designing retrofit strategies to meet code-based specifications is relatively simple as they require satisfying a pre-defined CDR_{LS} (e.g., 100%), which can be directly quantified and accomplished during the design process. However, optimising the retrofit design to achieve risk-oriented objectives (e.g., a target reduction in a seismic consequence of interest or life-cycle cost) can be more complex as the CDR_{LS} level at which the retrofitted structure meets these targets is unknown. Therefore, an iterative design procedure becomes necessary, involving the evaluation of numerous retrofit layouts/configurations and the calculation of the relevant metrics for optimisation until the desired target is realised. This process can be streamlined by characterising the relationship between CDR_{LS} and the metric chosen for optimisation, similar to the plots in Sections 3.2–3.4. This allows one to identify the exact level of CDR_{LS} corresponding to the metric's target value. One can then design and select a retrofit layout that upgrades the as-built structure to satisfy this CDR_{LS} level.

For instance, the relationship between CDR_{LS} and repair costs depicted in Figure 9a-c shows a decay curve. Initially, repair costs decrease sharply with incremental retrofitting but then level off. This pattern arises because retrofit schemes at high CDR_{LS} levels shift towards minor ductility enhancements (e.g., additional FRP layers or confinement), rather than strengthening new structural elements. This results in only minor improvements in seismic performance, particularly at DS1 and DS2. Similar patterns are observed in Figure 11a-c for the carbon emissions of repairing seismic damage.

The primary challenge, however, lies in generating sufficient data points to accurately fit a model describing the relationship of CDR_{LS} versus repair costs (or any metric of interest). In fact, each data point represents a particular scenario (retrofitted or as-built) with distinct CDR_{LS} and repair cost (or any metric) values. These points require the following information: (1) fragility parameters (μ_{DSi} and β); (2) hazard curves; and (3) consequence models (e.g., DLRs). Hazard curves are site-specific and can be easily acquired from several open-source tools.[1,2] Consequence models are widely available in the literature as well [31, 61]. Deriving fragility parameters, on the other hand, might be cumbersome as it often requires modelling and analysing the building's non-linear dynamic response under a range of ground-shaking scenarios.

To address this computational challenge, a practical approach is proposed. It leverages earlier work by the authors [22], which revealed that the relationship between CDR_{LS} and median fragility, μ_{DSi}, for incrementally retrofitted RC frames is pseudo-linear according to physics-based considerations. This means that only three points are necessary to fit the lines describing CDR_{LS} versus μ_{DSi} for each DS: one point representing the as-built structure, another reflecting a retrofit configuration with a significantly larger CDR_{LS}; and a third point with an intermediate CDR_{LS}. This approach is demonstrated in Figure 15a,b for both FRP wrapping (at CDR_{LS} of 42%, 80%, and 110%) and steel jacketing (at CDR_{LS} of 42%, 115%, and 147%), where the lines of CDR_{LS} versus μ_{DSi} fitted using three points are almost identical to those fitted using the entire points.

The lines derived in Figure 15a,b allows one to infer new CDR_{LS} versus μ_{DSi} points, eliminating the need for further modelling and non-linear analyses. These extrapolated points, combined with the original three points, can be then used to calculate any consequence metric of interest as outlined in Section 2. For example, Figure 15b,c presents CDR_{LS} values and their corresponding repair costs for the FRP wrapping and indirect losses

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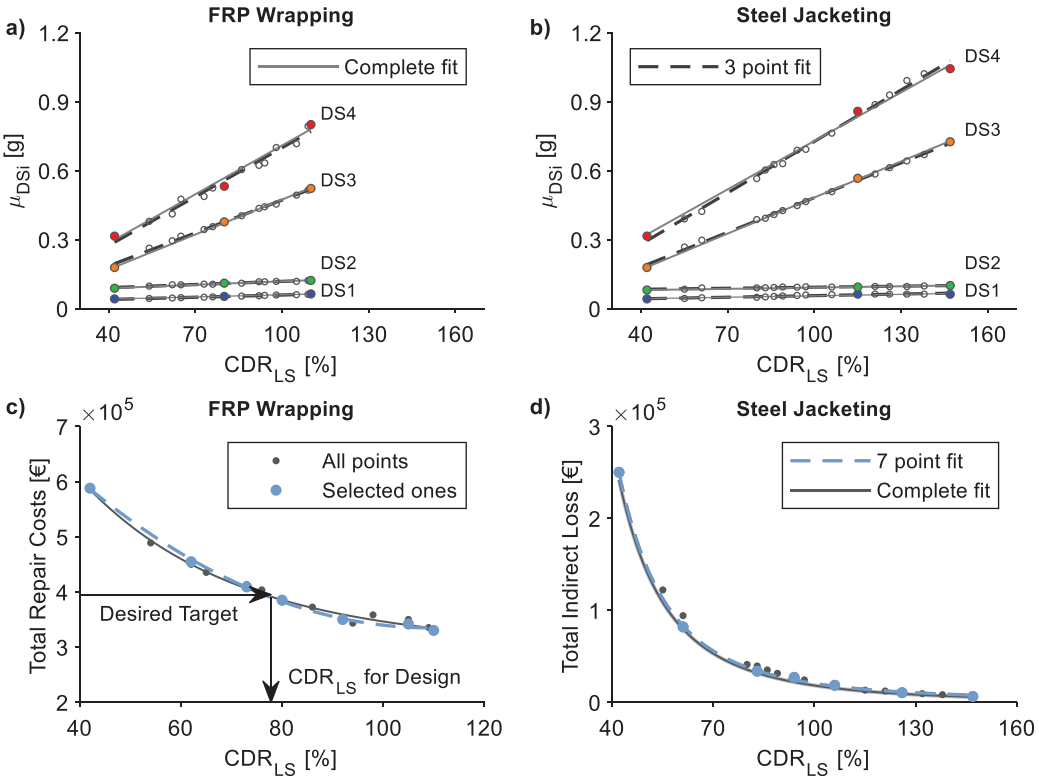


FIGURE 15 | (a, b) Linear CDR_{LS} versus μ_{DSi} relationship fitted with three points and with the entire point set for FRP wrapping and steel jacketing; (c, d) CDR_{LS} versus repair costs for FRP wrapping; and (d) CDR_{LS} versus indirect losses for steel jacketing.

for steel jacketing, respectively, assuming T of 50 years. Quadratic curves are fitted as follows: the ones with continuous lines are derived using the full data points, while the dashed curves use only seven points (three original and four extrapolated through the simplified linear model of CDR_{LS} vs. μ_{DSi}). Both curves align closely. It should be noted that other function types (e.g., power, exponential) are also applicable, provided there are sufficient data points. The curves in Figure 15c,d can guide retrofit design optimisation: a user may specify a target level of repair costs and then find the corresponding CDR_{LS} that the retrofitted structure must achieve to meet this target (as shown by the arrows). It is possible to implement the curves as well for estimating repair costs for any retrofitted structure if its CDR_{LS} is known.

It is worth noting that the proposed design simplification is broadly applicable to other RC frames as long as CDR_{LS} is displacement-based, and the retrofit strategy does not drastically alter the lateral stiffness. In such cases, the regression parameters (a and b) of the power-law model ($MIDR = aIM^b$), which is required to probabilistically characterise seismic demand and derive fragility relationships, will remain similar post-retrofitting. This ensures that improvements in the structure's displacement capacity cause a linear increase in μ_{DSi}, as demonstrated by Aljawhari et al. [22]. Nevertheless, applications to other structural systems and/or retrofit strategies should be verified on a case-by-case basis.

5 | Concluding Remarks

This study assessed the influence of non-conventional earthquake metrics on the decision-making and selection of suitable retrofit strategies, compared to traditional metrics, such as repair and life-cycle costs. The new metrics included life-cycle sustainability, covering the carbon generated by both the repair of seismic damage and retrofit interventions. Indirect losses incurred due to downtime were also examined by quantifying their risk-reduction benefit on life-cycle costs. The study considered a case study of an older, non-ductile RC frame, retrofitted using three common strategies: (1) FRP wrapping of joints and columns; (2) steel jacketing; (3) RC jacketing of columns. Numerous configurations for each strategy were analysed to address the variation in design choices and reflect improvement in seismic performance due to incremental retrofitting, measured by calculating capacity-to-life safety demand ratio, CDR_{LS}.

The study first evaluated life-cycle costs, including the cost of retrofit intervention and seismic damage repair, for all configurations. Simplified formulas to estimate the retrofit intervention cost based on its geometry were further derived. The computed life-cycle costs were then plotted as a function of CDR_{LS}. It was observed that the optimal CDR_{LS} minimising life-cycle costs varied remarkably by retrofit strategy: 95% for FRP wrapping; 117% and 146% for the steel and RC jacketing, respectively. This indicates that satisfying the code-based performance (CDR_{LS} = 100%) may not always result in an optimal retrofit solution. Amongst three strategies, RC jacketing outperformed the others in reducing life-cycle costs, regardless of CDR_{LS}. When indirect losses were incorporated in life-cycle costs, they did not influence the optimal CDR_{LS}, but they altered decision outcomes. At CDR_{LS} levels < 80%, FRP wrapping and steel jacketing proved superior due to their greater enhancement of the building's deformation capacity, which effectively reduced

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indirect losses. Indirect losses also played a critical role in assessing retrofit feasibility in low-hazard areas.

Next, this study considered sustainability by evaluating life-cycle carbon emissions, accounting for the emissions of seismic damage and those related to retrofit intervention, as a function of CDR_{LS}. Results revealed that life-cycle carbon did not follow the same trend as life-cycle costs, suggesting that the strategy with the least life-cycle carbon is not always the most sustainable, despite the strong between repair costs and carbon emissions of seismic damage. For example, FRP wrapping and steel jacketing at $CDR_{LS} < 80\%$ provided greater reductions in life-cycle carbon, as opposed to life-cycle costs, where RC jacketing was the superior strategy. Carbon emissions were also monetised via carbon taxes as an alternative way to integrate sustainability and life-cycle costs. The tax reduced the order of magnitude of carbon emissions and made their influence in retrofit decisions negligible when comparing different strategies.

It is essential to clarify that all types of seismic consequences were simply computed in this work at the building's (global) level, making the results more valid for assessing representative archetype building classes for regional seismic risk assessments. Other detailed approaches, like the FEMA P-58 component-based loss analysis [62], become more desirable when looking at individual buildings.

Finally, a practical approach for guiding retrofit design and optimisation was proposed. It begins by characterising the linear relationship between CDR_{LS} and median fragility, μ_{DSi}, using solely three data points (i.e., as-built, and two random retrofit layouts), as per Aljawhari et al. [22]. The linear relationship enables an easy extrapolation of additional CDR_{LS} versus μ_{DSi} points without further modelling or analyses. Both the original and extrapolated points are then utilised to compute any consequence metric (e.g., repair costs, carbon) and fit a function that relates the metric to CDR_{LS}. An example was shown featuring the case-study frame retrofitted with FRP wrapping and steel jacketing. The resulting function can be used by specifying the desired post-retrofit repair costs and acquiring the target CDR_{LS} that must be achieved in the retrofit design [63–65].

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

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